

Statistical Analysis

Descriptive statistical analysis was performed to summarize the numerical characteristics of the dataset. The following statistical measures were calculated:

- Mean
- Median
- Standard Deviation
- Minimum Value
- Maximum Value
- Quartiles

These statistics provide an overview of rainfall variability and help identify potential data inconsistencies before model training.

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# =====  
# FLOOD PREDICTION – EARLY WARNING SYSTEM  
# Step 1: Importing Libraries  
# =====  
  
# — Numerical & Data —  
import numpy as np  
import pandas as pd  
  
# — Visualisation —  
import matplotlib.pyplot as plt  
import matplotlib.ticker as mticker  
import seaborn as sns  
  
# — Scikit-learn: Preprocessing —  
from sklearn.model_selection import train_test_split  
from sklearn.preprocessing import StandardScaler  
  
# — Scikit-learn: Models —  
from sklearn.tree import DecisionTreeClassifier  
from sklearn.ensemble import RandomForestClassifier  
from sklearn.neighbors import KNeighborsClassifier  
  
# — XGBoost —  
from xgboost import XGBClassifier  
  
# — Evaluation —  
from sklearn.metrics import (accuracy_score, classification_report,  
                             confusion_matrix, roc_auc_score, roc_curve)
```

Data visualization and analysis played a crucial role in understanding the historical rainfall dataset used for flood prediction. Statistical summaries, histograms, count plots, box plots, correlation heatmaps, and pair plots were employed to explore rainfall distributions, identify outliers, analyze seasonal patterns, and examine relationships among meteorological variables. The analysis revealed that **June–September rainfall** exhibited the strongest influence on flood occurrence. These insights supported feature selection, improved preprocessing, and contributed to the development of an accurate machine learning model capable of achieving high flood prediction performance.